S4-1

The acoustic adaptation hypothesis across terrestrial vertebrates: a meta-analysis

Bárbara Freitas, Pietro B. D'Amelio, Borja Milá, Christophe Thébaud, Tim Janicke

Affiliation of presenting author:

National Museum of Natural Sciences, Spanish National Research Council (CSIC), Madrid, Spain Centre de Recherche sur la Biodiversité et l'Environnement, Toulouse, France

The design of acoustic traits does not only evolve in response to the interaction between signaler and receiver but is also shaped by the acoustic characteristics of the environment. Specifically, the acoustic adaptation hypothesis (AAH) predicts that the vegetation structure of the habitat determines frequency and temporal patterns of acoustic signals based on acoustic degradation and attenuation properties of vegetation. However, this hypothesis received mixed support depending on the studied taxonomic group and the methodology used. To test if the AAH is universally valid, we used a meta-analytic approach to synthesize 371 effect sizes from 75 studies and 57 taxa encompassing birds, amphibians and mammals. Overall, our results do not provide consistent support for the AAH, neither in within-species comparisons nor in among-species comparisons. However, when focusing only on birds, we found weak support for the AAH for intraspecies comparisons, suggesting some degree of adaptive phenotypic plastic response of acoustic traits to the habitat. For interspecific comparisons within birds, we also found support for the AAH, but this effect was not significant after excluding comparative studies that did not correct for their phylogenetic non-independence. Our synthesis suggests that vegetation structure does not universally shape acoustic signal evolution. We highlight the need for more empirical work on underrepresented taxa and advocate for a more detailed and quantitative characterisation of habitats to identify frequencies with the highest detection probability and determine if they are preferentially used. We stress that empirical tests of the AAH should focus on signals selected for long-range transmission.

S4-2

Genetic and environmental determinants of vocal frequency variation in tinkerbirds

Alexander Kirschel, Sifiso Lukhele, Anastasia Poyiadji, MaFeo Sebastianelli, Shannon Conradie, Stacey de Souza, Colleen Downs, Ara Monadjem, Bridgett vonHoldt.

Affiliation of presenting author:

Department of Biological Sciences, University of Cyprus, Cyprus

Vocal frequency in animals is shaped by a variety of intrinsic and extrinsic factors. A role of the environment in driving acoustic adaption is widely supported. Climate may also shape frequency both directly, via physiological effects on the vocal tract, but also indirectly through morphological adaptation to climate and temperature- dependent physiological constraints on song frequency. But much variation in vocal frequency has evolved over time between related species that cannot be attributed to environmental effects. Determining the relative effects of genetic and environmental factors and their interactions on vocal frequencies is a complex undertaking hitherto seldom attempted. Here, we assess the relative effects of body mass and bill length, abiotic and biotic environmental factors, sex, and genome-wide ancestry on song frequency in tinkerbirds across a hybrid zone in Southern Africa. Using path analysis, we determine the extent of direct and indirect effects of the environment and ancestry on vocal frequency. Our results confirm the strong negative effect of body mass on song frequency as well as the indirect effect of temperature via its effect on body mass. We also find that ancestry and sex affect song frequency. Yet we found little evidence for any acoustic adaptation to the vegetation structure of the environment. Overall, our study has shown that temperature-associated variation in morphology in accordance with Bergmann's rule directly affects vocal frequency in birds, but there is also a genetic component that explains further variation among species.

S4-3

Smog Songs

Sue Anne Zollinger

Affiliation of presenting author:

Manchester Metropolitan University, Department of Natural Sciences, Manchester M1 5GD, United Kingdom

Over the past 20 years, studies across the globe have begun to build a picture of how anthropogenic pollutants and environmental alterations are changing the behaviour, physiology, and fitness of wildlife. Much of this work has focused on the noise produced by human activities, and the auditory and non-auditory effects noise has on exposed animals, such as passerine songbirds. These studies have found that songbird in noisy habitats have often quite profound differences in their vocal behaviour compared with conspecifics in quieter areas, with evidence across many species for change in frequency, amplitude, timing, redundancy, or acoustic complexity. However, controlled lab studies have often failed to replicate many of the observed vocal differences between noise polluted and quieter habitats. One explanation for this might be that noise pollution often co-varies with other forms of anthropogenic disturbance, such air pollution emitted by traffic or industry. While we know that human singers' voices can be negatively impacted by air pollution either through short term effects due to irritation or inflammation to more lasting impairments of lung capacity or respiratory function, very little is known about the impacts of air quality on vocal behaviour and production in birds. Here I will present preliminary data on the deposition of airborne particulate matter in the lungs of songbirds, and links between air quality and song features in birdsong across an air pollution gradient, discussing potential physiological implications of smog on vocal production and quality.

S4-4

Changes in whistle production patterns of long-beaked common dolphins (*Delphinus capensis*) in controlled exposure experiments

Julie N. Oswald, Christina Day-Reader, Jasmine Lees, Caroline Casey, Selene Fregosi, Brandon Southall, Vincent M. Janik

Affiliation of presenting author:

Scottish Oceans Institute, University of St Andrews, Scotland

Many delphinid populations are exposed to Navy sonar signals, but little is known about the impacts of sonar on dolphin acoustic behaviour. We investigated the whistling behaviour of long-beaked common dolphins (Delphinus capensis) before, during and after exposure to mid-frequency active sonar during controlled exposure experiments (CEE; n = 5 controls, 14 treatments). Each CEE consisted of three 10-minute phases: pre-, during, and post-exposure. A total of 24 sonar pings were emitted in each exposure. To examine whistle use, 1324 whistle contours were traced across all conditions and categorised into types by an artificial neural network in ARTwarp. Of the resulting 204 whistle types, 96 were produced in all 3 experiment phases, and 30 were produced in only 1 phase (11 pre-exposure, 11 during, 8 post-exposure). Diversity turnover analysis to quantify changes in the types of whistles produced in the three phases showed that the whistle repertoire changed with experimental phase. Changes were greatest before vs during exposure for both treatments (angular turnover, dcos = 0.72) and controls (dcos = 0.46), with the largest turnover seen during sonar treatments. Of the five acoustic parameters measured from whistle contours, minimum frequency and mean slope were significantly lower during vs pre-exposure ($p<0.01$), and the number of frequency steps increased significantly during exposure ($p=0.006$) and then significantly decreased again post-exposure ($p<0.001$). These results show that common dolphins continue to communicate during sonar exposure but change whistle production patterns, presumably to maintain contact and coordinate movements in the presence of an unexpected stimulus.

S4-5

Soft yet complex: Acoustic trade-offs and social modulation of songs in star finches

Hana Goto, Henrik Brumm, Masayo Soma

Affiliation of presenting author:

Graduate School of Life Science, Hokkaido University, Japan

Song amplitude plays a crucial role in avian communication. As an advertisement signal, louder songs should have numerous merits. In contrast, limiting factors or selective forces against high song amplitude remain unexplored. To understand the evolution of soft songs, we have been investigating the singing behaviour of star finches (Bathilda ruficauda). Based on our preliminary finding that their song syllables are often complex (e.g. frequency modulation, two-voices, and possible nonlinear phenomena) and very low in d amplitude, we hypothesised a trade-off between amplitude and acoustic complexity of song syllables in this species. Supporting this idea, we found that both Wiener entropy and frequency bandwidth, used as measures of syllable acoustic complexity, were negatively correlated with vocal amplitude. Building on this, we then compared male song amplitude in different social contexts: songs directed at a female or produced in isolation. In other related species like zebra finches (Taeniopygia guttata), males increase song amplitude when they sing towards females. Zebra or star finches sing songs only for close-distance (side-by-side) courtship but not for territory defense. The question here is whether star finches can increase song amplitude, even when they use their extremely soft songs. As a result, we found that the males slightly but significantly increased amplitudes in female-directed contexts, possibly showing the performance close to their production limit. We will discuss possible factors that cause inter-individual or inter-syllable variations in amplitude increase in different social contexts.

S4-6

When song frequency meets identity information. A case study in the Ortolan Bunting

Lia Zampa, Paweł Szymański, Katarzyna Łosak, Tomasz S. Osiejuk

Affiliation of presenting author:

Department of Behavioural Ecology, Institute of Environmental Biology, Faculty of Biology, Adam Mickiewicz University, Poznań, Poland

Individually distinctive vocalizations play a key role in recognition across many taxa, yet the mechanisms shaping vocal identity remain debated. Is it merely a byproduct of morphological constraints and vocal learning, or is it actively selected for social and sexual advantages? To address this, we investigated vocal identity in the Ortolan Bunting (Emberiza hortulana), a territorial song-learning species with a small repertoire. Using spatial, vocal, and morphological data, we examined where, how, and why identity information is encoded. Our results reveal that individual signatures are stable across years and consistently present across all song types, song components, and song-sharing patterns, reinforcing that distinctiveness is not exclusive to socially complex systems. Males exhibited stronger individuality when surrounded by more competitors, supporting the role of intra-sexual selection in shaping vocal identity. Interestingly, song frequency, which differentiates shared song types among males and is crucial for encoding identity information in this species, does not correlate with body size. This suggests that frequency-based identity cues are not a byproduct of morphological constraints but are instead shaped by vocal learning or selection for individuality. Although frequency did not reliably signal body size, playback experiments showed males adjusted their responses based on the intruder's song frequency. This suggests that frequency remains relevant in social interactions, potentially influencing perceived threats or competitive dynamics, though the exact mechanism remains unclear. These findings provide novel insights on how territorial dynamics, learning processes, and morphological constraints interact to shape vocal identity and influence recognition systems.